A low-threat practice-room interface for neurodivergent game learning

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Fast multiplayer games can convert practice into threat when social judgment, sensory load, ambiguous damage, and repeated failure are experienced as one continuous demand. This manuscript reports a single-user design study of league.aolabs.io, a public review room for League of Legends practice. The system does not claim clinical efficacy, population generality, or competitive optimization. It translates a user-provided learning arc into a low-threat review surface: three champion pages, source-bounded recording review, archived situation notes, public logs, and a downloadable record. The strongest contribution is a design pattern rather than a gameplay result: make the next repetition easier to start by keeping the active champions visible, putting recording feedback first, encoding coach intent directly in the paragraph color system, and moving older situation material into an archive. The interface treats bots as a practice room, human games as performance, death-with-intent as a valid rep, and recording review as mental practice. This bounded case suggests how personal learning software can support high-repetition play without becoming another source of evaluation pressure.

1 Introduction

Expert performance depends on repeated, structured practice, but practice is only useful when the learner can remain in the task long enough for repetitions to accumulate (*1*). In competitive games, especially player-versus-player games, the immediate source of failure is not only mechanical difficulty. The learner must also manage uncertainty, sensory load, social interpretation, and the felt cost of visible mistakes. Anxiety can impair attentional control (*2*); cognitive load can interfere with learning when too many elements must be processed at once (*3*). A player who is trying to learn a champion, lane, camera, item build, opponent behavior, teammate expectations, and self-regulation at the same time may experience the game as a judgment environment rather than a practice environment.

The system described here began from a simple transfer from music practice: keep the task playable, choose one thing, trust repetitions, and let improvement compound. The same principle was applied to League of Legends. The first rule was not to win lane or outplay an opponent; it was to farm and survive. That rule later compressed into the phrase "next safe minion." As the practice record grew, the problem became less about collecting more advice and more about making advice reviewable without overwhelming the player before queueing.

league.aolabs.io is a public AO Labs page built for that purpose. It is not a Riot Games product, a general League guide, or an official ranked-MMR system. It is a three-champion review room: Samira and Cait have source-confirmed recording pages, while Fizz has a clean empty recording page until Fizz footage exists. The design is source-bounded: it uses the user's supplied chat arc, public notes, local recordings, and local client reads as its corpus, and it avoids claims that would require clinical, population-scale, or competitive outcome evidence.

2 Results

2.1 A long learning arc was reduced to a small review surface

The seed corpus contained a long narrative account of League practice. The arc included lane fear, camera stability, sensory control, champion feel, bot practice, death tolerance, public yapping as pressure regulation, and champion-specific rules for Samira, Caitlyn, and Fizz. The current design result is not a long visible essay, a motivational hero, or a broad champion library. The page exposes Samira, Caitlyn, and Fizz at the top, puts recording review immediately after those champion cards, and moves older situation notes into an archive.

This reduction is the main interface result. It makes the review surface compatible with the user's stated constraint: the page should be pretty, clean, non-poster-like, and not overwhelming. The long record remains useful as paper, log, and archive context, but the first screen does not ask the user to process a giant review tray, a generic champion taxonomy, or stale situation cards before finding the current recording feedback.

Table 1. Compression of the source arc into champion-specific review units.

Source burden	Review unit	Interface expression
Enemy lane feels like judgment	Archived situation	Next safe minion remains in the paper record
Damage triggers panic	Mindset situation	Damage is information
Samira has many buttons	Button-order situation	Q before E; W saved for real danger
Low targets bait chase	Chase situation	Low plus reachable, not low plus fog
Champion fit affects repetition	Three champion pages	Samira and Cait recordings; Fizz empty until footage exists

2.2 The interface itself became part of the research record

The public page is not a neutral wrapper around the recordings. It is part of the intervention being studied. For that reason, the paper records the actual interface, not only the backend, manifest, or coaching text. The live page begins with the shared AO Labs header, a quiet paper strip, three champion cards, and then the current selected-champion review state. Samira, Caitlyn, and Fizz use exact user-supplied photographs: the hat photo for Samira, the white-top bed photo for Caitlyn, and the peace-sign photo for Fizz. The Caitlyn card is visually framed from the top-right so any card crop comes from the bottom or left while the served asset remains the original uploaded image. This matters because the page is meant to feel like a personal practice room rather than a generic analytics product.

The top review tile combines the selected champion's rank trend plot and latest synced state into one calm surface. The plot uses exact per-game rank equivalents rather than broad ranges, and the state paragraph is latest-weighted: the newest synced full game carries the coaching state, the last three full games set the current pattern when enough footage exists, and the older archive median remains baseline context only. It is not live match telemetry; it is current through the latest generated recordings manifest. That top tile is intentionally not a broad dashboard. It exists to answer one question before the next game: what level is the latest synced play reading as, and what current pattern is most worth fixing next?

Each recording row is also a designed reading object. The title names the main pattern, the first line gives queue, champion, time, length, rank read, and compact K/D/A/CS context, and the paragraph carries the actual coaching. The color semantics are now part of the product contract: green marks what was good and should be kept, red marks what was bad and should stop, pink marks the exact next-game action, and neutral dark text carries the timestamps, evidence, stats, and context that make the highlighted claims understandable. The pink span is reserved for action text that remains useful when read alone, so a timestamped setup can stay neutral while the instruction itself is visually distinct. This avoids making Alan infer a hidden legend from several arbitrary colors.

Table 2. Current interface elements and the paper obligation they create.

Visible surface	Function in the practice room	Paper record required
Shared AO Labs header and paper strip	Places the tool inside the AO Labs suite and keeps the PDF reachable without adding navigation clutter	Name the public route, suite context, and paper access pattern
Samira, Caitlyn, and Fizz cards	Keep the active champion set small and personal; avoid a broad champion library	State why only these three champions are exposed, which supplied photos are exact, and why empty champion pages stay footage-gated
Rank trend and latest synced state tile	Shows the latest synced rank-equivalent read and newest-game blocker before the recording list	Describe the rank field, aggregation, freshness boundary, and state-generation rule
Post-game queue	Shows whether capture, processing, publishing, and posting are waiting, active, or complete	Describe queue ordering, ETA role, one-at-a-time processing, and lag-reduction constraints
Recording card	Couples video, timestamped review, compact stats, and action script in one row	Describe the card anatomy and how evidence, failure lanes, and timestamps are rendered
Highlighted coach clauses	Turns one paragraph into a readable coach key without adding panels	Record the reading model: green keep, red stop, pink next action, neutral support text
Archive and public log	Preserve older situation rules and public notes without competing with current review	State which material is demoted below the active recordings and why

2.3 Recordings replaced situations as the primary review unit

The site now organizes the visible page by recordings rather than by abstract lesson cards. Situation notes still exist because they explain the origin of the rules, but they are archived below the current recording review. Selecting Samira shows recording rows with the video, game type, champion, match time, game length, one title, and one compact game-story paragraph. Selecting Fizz shows the same page structure with no fake feedback, because no Fizz footage has been source-confirmed. This format was chosen because the user needed enough detail to recognize what happened in the actual game without reading a giant rule wall, a stat audit, or a card taxonomy.

Table 3. Example Samira rules retained in the archived situation record.

Situation prompt	Combined response
The fight feels urgent and E wants to become the first button.	Start with Q or an auto so the fight gives information before the dash commits the body.
The enemy has CC that is not recognized yet.	Stay outside on Q-only, watch one danger spell, use W or sidestep for visible projectiles, and E only after the scary spell is gone.
A low-health enemy runs toward fog or teammates.	Stop at the edge; low only matters when the target is also reachable and cheap to kill.
W feels necessary immediately after dashing in.	Hold W for the real spell; it is a parry for visible danger, not armor against being close.
R works and the excitement makes staying feel automatic.	Treat R as the reward, then leave unless the next target is already low and reachable.
E resets after a takedown.	Read the reset as a new check, not an instruction to dash into tank, tower, fog, or fresh crowd control.
One hit lands and feels like proof that the enemy is competent.	Treat damage as information, step back, stabilize the camera, and choose the next action.
The player is near a team fight, R is not ready, and waiting there costs HP.	Move to the edge, build style from safety, and enter only when S rank exists and the fight is already committed.
The player tunnels on one low target.	Check target, danger, target, danger; stop chasing if the chaser becomes the low target.

2.4 Champion focus was narrowed to Samira, Caitlyn, and Fizz

The visible champion set is no longer a broad library or a tier list. It is deliberately limited to Samira, Caitlyn, and Fizz. All three cards use exact user-supplied photos rather than approximate K-pop stand-ins, and Caitlyn is exposed as Cait in the interface because the user started a Cait game and uses that name during play. Samira remains a recorded focus because she provides a high-reward dash-reset-R loop, but she also creates the central panic trap of using E as an anxiety button. Caitlyn is the comfort sniper page for range, farming, and unavailable positioning, with footage-gated review once Cait recordings exist. Fizz remains the comparison page because his button shape makes the desired loop clear: mark, enter, stab, dodge, leave.

The older champion-fit material remains useful as source history, but it is not allowed to compete with the current practice page. The public page uses the visible champion cards as the selection mechanism, keeps recording feedback first, and lets Caitlyn or Fizz stay empty until the recording folder contains footage for that champion. This avoids fake specificity and preserves the rule that no footage means no feedback.

2.5 Raw recordings converted carry games into timestamped improvement lanes

On May 18–24, 2026, the public page added a generated recordings manifest from the local League of Legends Highlights folder and the live capture publisher. The current manifest contains 50 recordings across 38 matches and includes short highlight clips plus longer review recordings. Champion detection is bounded to source evidence: older rows remain Samira when sampled frames show Samira in the team list or center the Soul Fighter Samira model and nameplate, while the newest Cait row is source-confirmed from the local client champion id and visible Cait review footage. No champion label is treated as source-confirmed without that kind of evidence.

The recording review does not attempt to become a full coaching dashboard. The public surface now keeps champion cards first, then places one combined rank-and-state tile with a compact exact-rank trend plot and current-state coaching paragraph for the selected champion. That paragraph is generated from the current manifest on every sync rather than preserved as a static coaching slogan: it reports the newest exact rank read, compact K/D/A and CS context when available, the newest-game blocker, and one next-game action from the newest review. Each row below uses one title followed by one narrative paragraph that folds in queue type, champion, match time, rounded game length, compact K/D/A and CS context when the local client provides it, visible evidence, direct mistake/fix coaching, failure-chain proof, multiple mistake-type lanes, one honest positive note when the footage supports it, one separate second focus for an easy next-game habit, and one exact most-likely rank-equivalent macro read for full-review games. The rank estimate uses Riot's current tier structure and queue distinctions (*4*, *5*), then gates confidence with evidence from full-game review, client K/D/A and CS, visible timestamps, queue pressure, and cross-game calibration. The line plot uses the same per-recording exact-rank field as the visible row label, so a game cannot appear as a broad range in one place and a different rank in another. That gate is deliberately conservative: expert-play research supports repeated, consistent ranked practice as part of high-skill evidence (*6*), self-monitoring through CS, gold, and post-game statistics as a useful review signal (*7*), and objective, movement, and vision behavior as rank-relevant competitive evidence (*8*). Co-op vs. AI games can therefore produce a useful macro-equivalent read, but high-confidence ranked-transfer evidence is reserved for PvP or ranked full games; current ranked rows are treated separately from older bot-practice rows. Rows are sorted with the newest match first and, within a match, the longest recording first. The paragraph is intentionally bounded: it names the notable visible game moments in story form, uses timestamps only as evidence anchors, shows K/D/A and CS as compact stat context rather than a separate stat dashboard, weaves red constructive-criticism phrases, green praise phrases, failure evidence, mistake-type phrases, and the second-focus sentence into one story instead of adding separate panels, and turns visible game-clock timestamps into inline controls that seek and play the matching video from that moment. After the May 23 evidence-lane fix, new automatic full-game reviews must scan all visible aspects that can honestly be judged, including macro, reset timing, objective conversion, camera stability, spacing, entry timing, target choice, cooldown and crowd-control accounting, wave state, fog/vision pathing, and repeated cursor/pathing habits. They must include a verified game-clock timestamp for the beginning or nearest visible beginning of the main mistake window, name the leak or consequence, include visible event evidence, include failure-chain proof, list at least three distinct mistake lanes when the footage supports them, match visible timestamps to verified clock anchors, explain why the recommended action is correct, include one distinct secondary improvement lane, give a timestamped replacement action script for the main mistake instead of only naming a category such as group, reset, pressure mid, spacing, or target selection, and avoid naming allied or enemy champions or exact jungle buffs unless the roster/nameplate or camp label is verified. The latest branch-standard update also forbids abstract cash-out wording in full-game rows: the main timestamp must describe the visible tower, wave, ally-front, enemy-visibility, and K/D/A/CS context, then separate the playable options as free tower, safe blocker, wave then recall, or leave. If the detailed model pass fails or returns no usable JSON, the sync now builds a conservative deterministic fallback from verified clock frames and still audits that fallback against the same visible-feedback standard before publication, so a missing model response does not create a publish-blocked state by itself. The current Cait target is damage-without-death conversion: keep Cait unavailable after the useful damage window, then choose only a safe wave, protected objective hit, tower defense, or reset before a forward click can turn into another death.

The May 24 example-review update makes the row contract stricter: every new current-standard full-game row carries a source-backed victory or defeat label when the League Client match history provides it, uses one sharp mistake title, and starts the review with one natural key timestamp sentence that names the exact visible state and translates the idea into a literal next click. Red text is reserved for the leak, green text for the habit to keep, pink text for one next-game *Rep:* sentence, and neutral text supplies timestamped evidence and K/D/A–CS context. Rows are rejected when they fall back to generic review-frame language, label lists such as “mistake category” and “correct next click,” or abstract branch terms that do not become an if-this-then-that click rule.

2.6 The public log made mindset part of the artifact

The public log starts with source entries for the seed arc import, the practice-room frame, the current Samira law, and later user-supplied Samira proof-game notes. This is intentionally public. The design choice is that thoughts, panic notes, and mindset rules are not private by default; they are part of the learning artifact. The only excluded material is unrelated sensitive information such as credentials, access tokens, or private third-party identifiers.

3 Discussion

The case suggests that the useful interface for some high-repetition learners is not less information in the abstract, but better placement of information. A long learning conversation can contain many correct rules while still being hard to use before a game. The revised review room limits the first screen to Samira, Caitlyn, and Fizz, puts playable recordings before archived situation notes, and gives more detail only where footage exists. That is consistent with cognitive-load theory: reducing simultaneous elements can make practice more tractable (*3*). It is also compatible with mental practice research, because the page gives the user a concrete game moment to rehearse before performance (*9*).

The work also reframes death and failure. The site does not encourage intentional feeding or fake failure. It uses the rule “choose the fight; death is allowed; winning is intended.” That distinction matters. It keeps agency with the learner while preserving the real objective of the game. It also separates practice failure from collapse: a failed engage that tested damage, W timing, or exit timing is a rep; walking in only to die is not.

Several limits are important. First, this is a single-user artifact, not a clinical intervention. The user self-described ADHD/autism traits and supplied a practice narrative, but the site does not diagnose, treat, or generalize to neurodivergent players as a class. Second, the current evidence is artifact-level: source text, distilled rules, local recordings, local League Client match-history reads, a rendered page, and the paper itself. It does not include a controlled comparison or a ranked outcome. Third, the system deliberately avoids public Riot API import in version 1. That keeps the artifact focused on review and self-regulation rather than turning it into a general stat tracker.

4 Materials and Methods

Source corpus

The seed corpus was the user-provided League learning arc pasted into the implementation thread on May 17, 2026. It described the user's own League practice, including early Caitlyn farming goals, Samira drills, bot-ladder practice, camera and sensory adjustments, death exposure, and champion-fit comparisons. Later May 17 and May 18 prompts added Samira crowd-control, W-blocking, poke-lane, bait-recognition, team-fight, bad-team, normal-game, and proof-game refinements. A May 18 local recording folder then supplied 15 files across matches `NA1-5563247854`, `NA1-5563301586`, and `NA1-5563352800`. Local replay files supplied the match-time anchor, League client logs supplied queue id `880`, Riot's published queue table identifies `880` as Co-op vs. AI Beginner Bot games,(*10*) and the local League Client match-history endpoint supplied K/D/A, CS, and game duration when the client was available. The implementation treated these texts, recordings, replay files, local logs, and local client reads as the current source of truth for the public situation notes and recording review.

Interface requirements

The page was constrained by user instructions gathered during planning and revision: one page only, clean and pretty, same AO Labs vibe, no poster hero, no dense dashboard, no redundant helper labels, no overexplaining, public notes by default, one quiet paper strip, detailed coaching without a giant checklist, a readable queue, and a color system that does not require a legend. The page therefore uses the shared AO Labs header, warm off-white palette, dark ink text, 8 px cards, restrained borders and shadows, compact fragments, one visible PDF action, and a four-part coaching color key: green for the behavior to keep, red for the behavior to stop, pink for the next-game action, and neutral dark text for evidence, stats, timestamps, and context. The paragraph must still read naturally as coach prose; the colors are applied after the coaching is written so that skimming only green, only red, or only pink answers a different practical question. This interface description is treated as part of manuscript accuracy: when the page changes in a way that changes what Alan sees, how evidence is displayed, how queue state is communicated, or how coaching intent is encoded, the paper must be checked or updated in the same work cycle.

Rule extraction

Rules were extracted by finding repeated gameplay or mindset triggers in the source corpus and translating each into a one-sentence situation prompt plus one combined response paragraph. The extraction favored recognition over slogan compression: the prompt had to name the exact stressful moment, and the response had to combine action, reason, and practice rep without separate "do," "avoid," or "drill" labels. For example, the Samira section retained the underlying rules "Q first; earn the E," "W is parry, not armor," and "R is reward, not start," but the public card now starts from the situation that makes the rule necessary. The death section retained "choose the fight; death is allowed; winning is intended." The bot-practice section retained the practice ladder and the current-mode cue.

Public artifact implementation

The website is served by a small Node application on Railway. It serves the hand-authored page, generated recording manifest, exact champion-card photos, poster frames, and paper artifacts from the public directory and exposes `/api/logs` for public note reads and writes. Local recording media is served with byte-range responses so inline timestamp controls can seek directly into a video row. For manually reviewed long recordings, the manifest can store frame-derived game-clock anchors, so a narrative timestamp seeks by the visible in-game clock rather than by raw clip offset. Large recording videos are kept out of the Railway upload payload; when a local video file is not present in the deployment image, the server redirects that recording request to a GitHub-hosted media copy for the same repository path. A local sync script scans the source Highlights folder, reads replay and League client logs for match and queue metadata, reads local League Client match history for champion id, K/D/A, CS, and game duration when available, copies small `.webm` or `.mp4` files into the public recording directory, compresses large recordings into higher-bitrate deployable `.mp4` assets, extracts poster frames with `ffmpeg`, reuses cached feedback for unchanged files only when the visible-feedback standard still passes, computes a research-calibrated single-rank macro estimate for each full-review game, stores the exact rank and numeric rank value for trend plotting, regenerates the latest synced selected-champion state from the newest full-game review for that champion, stores per-game confidence signals and blockers, stores the external research basis used by the rubric, and refreshes the manifest that the browser loads. The high-confidence gate requires ranked or PvP full-game evidence; current ranked rows are eligible for high-confidence transfer reads while older Co-op vs. AI rows stay lower confidence. The same sync path audits new automatic full-game reviews for a verified game-clock timestamp on the main mistake window, a named decision leak, event evidence, matching clock anchors for any visible timestamp, a timestamped replacement action script, a plain-language reason for the recommended action, a distinct secondary improvement lane, and a source-backed wording gate for champion and jungle-camp names. When the detailed model pass fails, that audit no longer has to block publication solely because the model returned no JSON: the script selects the strongest verified visible clock moments, prefers later macro/objective windows when supported, writes a conservative mistake/fix paragraph, and labels the review limit as a deterministic fallback before the audit runs. The default 2 FPS capture can support macro, spacing, pathing, camera, entry, target-choice, and repeated cursor-pattern habits, but it cannot honestly support frame-perfect mechanics claims. The newest prompt explicitly allows direct Challenger-path critique while forbidding insults and unsupported certainty, tells the model not to assume mechanics are the blocker when visible coordination is fine and decision-making is the real leak, and requires unverified champion names to be replaced with role words such as ally, enemy, support, jungler, defender, or teammate. A separate timestamp and feedback audit rechecks the generated manifest before publication. A local live-recorder script can also watch for the League game process and local League Client gameflow state, capture the League window region at low priority, prefer low-impact hardware H.264 settings when available, split capture into short segments, fast-forward lower-motion segments, preserve higher-motion segments at normal speed, enqueue post-game reviews so one review is processed at a time while the next game can continue recording, and publish queue status with per-stage and ready-time estimates. The recorder now records an owner process identifier, exits future duplicate instances when they lose the lock, rejects stale queued status posts that do not come from the active recorder, and can run a local guard that strips stale in-memory queue rows when an older process cannot be terminated. By default, if the recorder restarts while a game is already running, it ignores that partial game and waits for the next full match; a recovery environment flag can capture a separate partial session instead. The recorder stores video and cursor motion, not raw keyboard text. The manifest records match time, queue type, champion, game duration, K/D/A, CS, recording duration, one takeaway, a compact recording-specific paragraph, secondary improvement focus, capture FPS, exact rank estimate for full-review rows, evidence basis, and review limit for each item, then sorts newest matches first and longest recordings first within each match. On Railway, persistent storage is provided through the configured volume path, and writes are gated by `LEAGUE_WRITE_TOKEN`. This keeps public notes browser-independent without adding a fake local-only note form or claiming public Riot API integration that is not present in version 1.

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Author contributions

Alan N. Pham supplied the practice corpus, product constraints, and acceptance criteria. The manuscript and site were assembled from those source materials in the AO Labs workspace.

Competing interests

The author owns and operates AO Labs. The site is a personal public artifact and is not endorsed by Riot Games.

Data availability

The first public data source is the League practice arc supplied in the implementation thread and the public log exposed through league.aolabs.io. The current public page also exposes the May 18–23, 2026 recording manifest for 48 recordings across 36 local match identifiers, poster frames, local queue metadata, local client champion id, K/D/A and CS when available, per-recording feedback, secondary improvement focus, research-calibrated rank-equivalent macro estimates for 34 full-review rows, one main queue focus, compact post-game queue status, and timestamp anchors that seek into the playable videos. The manifest also exposes a rank-calibration block with the research sources, high-confidence requirements, and confidence blockers that separate ranked/PvP rows from bot-practice rows. The newest automatic reviews were corrected after frame inspection showed that unsupported generic reset critiques and unverified entity labels should be replaced with timestamped evidence about shutdown-lead conversion, base re-entry sync, spend-after-base-payout decisions, queue/publish state, champion-name limits, jungle-camp label limits, K/D/A and CS context, and a second improvement lane that can be judged from the default 2 FPS capture. The latest publish reliability fix adds a deterministic timestamped fallback when the detailed model pass fails, so the page can still publish a bounded review that states its limit rather than staying blocked after a clip is created. Public Riot API match history is not imported in this version.

Code availability

The implementation is intended for the `nalalalan/league-app` repository, with the public site, PDF, manuscript source, and log API served at league.aolabs.io.

Additional information

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